

CLIENT BRIEF

Vari Smart Operations Platform

Deep Learning Subject Project + Hackathon MVP
Track 2: Crowd, Mobility & Resource Management

1. Executive Summary

Vari Smart Operations Platform is a unified AI-enabled operational management system for Vari organizers. It monitors crowd movement, predicts congestion, coordinates volunteers, monitors food/water resources, and raises sanitation alerts.

The system is designed as both a **hackathon MVP** and a **Deep Learning course project**. The main pipeline is: video/data input → deep learning inference → analytics → prediction/risk → recommendation → real-time dashboard.

2. Problem Statement Coverage

Track Area	MVP Capability	Priority
Smart Crowd Monitoring & Movement	Person detection/counting, density, risk, prediction, live map	Must Have
Smart Seva & Volunteer Coordination	Volunteer status and nearest suitable volunteer allocation	Must Have
Food & Water Distribution	Stock monitoring, shortage alerts and redistribution recommendation	Must Have
Waste & Sanitation	Waste level, cleaning alerts and task creation	Basic MVP

3. Project Vision

SENSE → PREDICT → OPTIMIZE → ACT

- **Sense:** collect crowd, volunteer, resource and sanitation information.
- **Predict:** estimate crowd growth and operational risk using AI/ML.
- **Optimize:** recommend volunteers, routes and resource transfers.
- **Act:** present clear recommendations to the operator.

4. Main MVP Features

- **Live Crowd Monitoring:** crowd count, capacity, density and risk by zone.
- **Crowd Risk Prediction:** predict crowd level for the next 15–30 minutes.
- **Live Vari Map:** display zones, gates, volunteers, resources and alerts.
- **Smart Volunteer Allocation:** identify and recommend nearby available volunteers.
- **Resource Monitoring:** food/water levels and shortage detection.
- **AI Recommendation Engine:** recommend practical actions before situations become critical.
- **Waste/Sanitation:** monitor waste zones and create cleaning alerts.

5. Deep Learning Component

The Deep Learning course component will be centered on computer vision. A pretrained **YOLO object-detection model** can detect people in crowd videos. OpenCV handles video processing.

Video / CCTV → YOLO Person Detection → Person Count → Crowd Density → Risk → Alert / Recommendation

- **Object Detection:** YOLO for person detection and counting.
- **Computer Vision:** OpenCV for frame processing and visualization.

- **Temporal Prediction:** optional LSTM/GRU model for crowd-count forecasting.
- **Transfer Learning:** optional fine-tuning if a suitable dataset is available.
- **Advanced extension:** abnormal crowd-flow/event detection.

Recommended course scope: demonstrate YOLO as the primary Deep Learning model and, if time permits, compare an LSTM/GRU prediction model with a simpler ML baseline.

6. Crowd Monitoring & Risk Logic

```
density (%) = (current people / zone capacity) × 100

< 60% → SAFE
60–85% → MODERATE
85–100% → HIGH
> 100% → CRITICAL
```

These thresholds are prototype rules for the MVP and should be calibrated with real operational data before any real-world deployment.

7. AI Decision Examples

- Gate 2 reaches 126% capacity → generate critical alert and recommend redirecting incoming crowd.
- Current crowd is 8,200 and predicted 15-minute crowd is 9,800 → warn the operator early.
- Critical zone requires 8 volunteers → find nearby available volunteers and recommend deployment.
- Water station falls below 25% → identify a nearby surplus station and recommend transfer.
- Sanitation zone exceeds 85% → create a cleaning task.

8. Dashboard

- Total monitored crowd
- Active alerts
- Available/on-duty volunteers
- Food and water availability
- Live Vari map
- Crowd density and prediction charts
- Alert severity
- AI recommendations
- Resource and sanitation status

9. System Architecture

```
React Dashboard
↓ REST + Socket.IO
Node.js + Express Backend
↓
MongoDB
↑↓
FastAPI AI/ML Service
↓
YOLO + OpenCV + Scikit-learn / Deep Learning Models
```

Node.js acts as the application/business layer. Python/FastAPI provides AI inference and model-serving endpoints.

10. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	Dashboard/UI
Charts	Recharts	Analytics
Map	Leaflet + OpenStreetMap	Operational map
Backend	Node.js + Express.js	REST APIs/business logic
Real-time	Socket.IO	Live updates/alerts
Database	MongoDB + Mongoose	Operational data
AI Service	Python + FastAPI	Model inference
Deep Learning	YOLO + OpenCV	Crowd/person detection
ML	scikit-learn	Prediction/baselines
Optional DL	LSTM/GRU	Time-series prediction
Deployment	Vercel + Render	MVP hosting

11. Data Strategy

Real event infrastructure may not be available during development, so the MVP uses a hybrid data strategy.

- Suitable crowd videos for demonstrating person detection.
- Simulated crowd, resource, volunteer and sanitation data.
- Historical/simulated time-series data for prediction experiments.
- A controlled simulation mode for the final hackathon demonstration.

Simulated values must be clearly identified as demo/simulation data and not presented as actual Vari measurements.

12. Volunteer Coordination

Critical Zone → Required Volunteers → Filter Available → Calculate Distance → Select Suitable Volunteers → Create Task

Volunteer records contain location, skills and availability. The MVP can use nearest-volunteer selection; more advanced versions can use optimization algorithms.

13. Food & Water Optimization

Shortage Station → Find Surplus Station → Calculate Distance → Recommend Transfer → Operator Action

For the MVP, rule-based redistribution is sufficient. Future versions can use linear programming or vehicle-routing optimization.

14. Waste & Sanitation

- Waste level percentage
- Last cleaning time
- Zone status
- Cleaning priority
- Assigned team
- Recommended collection route

15. Team Structure — 3 Members

Member	Primary Responsibility	Key Deliverables
Member 1	Frontend	React dashboard, map, charts, pages, real-time UI
Member 2	Backend	Express APIs, MongoDB, Socket.IO, alerts, allocation logic
Member 3	AI/ML	YOLO, OpenCV, prediction, recommendation engine, FastAPI

16. Five-Day Roadmap

Day	Focus	Expected Result
Day 1	Foundation	React, Express, MongoDB, YOLO setup and first detection
Day 2	Core Features	Map, charts, APIs, density and prediction
Day 3	Intelligence	Alerts, volunteer/resource allocation, recommendations
Day 4	Integration	React ↔ Node ↔ FastAPI ↔ MongoDB, simulation flow
Day 5	Polish & Demo	Testing, deployment, UI polish and presentation

17. Detailed Team Tasks

Member 1 — Frontend

- Day 1: React/Tailwind setup, dashboard shell, navigation, reusable cards.
- Day 2: Leaflet map, crowd markers, charts, crowd page.
- Day 3: Volunteer/resource/alert/recommendation pages and action buttons.
- Day 4: Axios + Socket.IO integration and end-to-end testing.
- Day 5: Responsive UI, animations, error states and final polish.

Member 2 — Backend

- Day 1: Express, MongoDB, Mongoose, Socket.IO and schemas.
- Day 2: Crowd, volunteer, resource and alert APIs.
- Day 3: Density/risk engine, real-time alerts, volunteer and resource allocation.
- Day 4: FastAPI integration, database integration and complete testing.
- Day 5: Deployment, environment variables, bug fixing.

Member 3 — AI/ML

- Day 1: Python, YOLO, OpenCV, sample video inference.
- Day 2: Density/risk calculation, data preparation and short-term prediction.
- Day 3: Recommendation engine, route/volunteer optimization and FastAPI endpoints.
- Day 4: Integrate FastAPI with Node.js and prepare fallback simulation.
- Day 5: AI debugging, inference optimization and final demo validation.

18. Core Database Collections

Collection	Important Fields
users	name, email, role, phone
volunteers	name, location, skills, status
crowdZones	name, capacity, currentCount, density, riskLevel, coordinates
resources	type, location, quantity, capacity, status

Collection	Important Fields
alerts	type, severity, location, message, status, createdAt
tasks	title, assignedVolunteer, location, priority, status

19. Core APIs

```
Crowd:
GET /api/crowd
GET /api/crowd/:zoneId
POST /api/crowd/update

Volunteers:
GET /api/volunteers
POST /api/volunteers
PUT /api/volunteers/:id

Resources:
GET /api/resources
PUT /api/resources/:id

Alerts:
GET /api/alerts
POST /api/alerts
PUT /api/alerts/:id

AI:
POST /detect-crowd
POST /predict-crowd
POST /recommend
POST /optimize-volunteers
```

20. Simulation Mode

A simulation mode is essential for a reliable hackathon demo because real CCTV, IoT sensors and operational systems will not be available.

```
SIMULATE CROWD SURGE

Gate 2:
72% → 86% → 103% → 128%

System response:
• Alert generated
• Prediction updated
• Volunteers recommended
• Alternate route recommended
```

21. Three-Minute Demo Flow

Time	Demo
0:00–0:30	Explain crowd, volunteer, resource and sanitation challenges.
0:30–1:00	Show dashboard, live map, crowd, volunteers, alerts and resources.
1:00–1:30	Run crowd video through YOLO and show count, density and risk.
1:30–2:00	Show prediction and recommended crowd/volunteer response.
2:00–2:30	Show water shortage and redistribution recommendation.
2:30–3:00	Show final state and explain Sense → Predict → Optimize → Act.

22. Repository Structure

```
vari-smart-ops/
├── frontend/
├── backend/
├── ai-service/
└── data/
```

23. Features to Avoid in the 4–5 Day MVP

- Native mobile app
- Complex authentication/role system
- Real IoT infrastructure
- Real government/event-system integration
- Face recognition
- Large custom DL model from scratch
- Production-grade navigation
- Complex chatbot
- Large manual data-entry system

24. MVP Success Criteria

- Dashboard loads realistic simulation data.
- Live map displays operational zones and resources.
- YOLO detects/counts people in demonstration video.
- Crowd density and risk update correctly.
- Short-term prediction is visible.
- Critical events generate real-time alerts.
- Nearby volunteers can be recommended/assigned.
- Food/water shortage creates redistribution recommendation.
- Sanitation issue creates a task/alert.
- AI recommendation panel explains the suggested action.
- Frontend, backend, database and AI service communicate successfully.
- Complete demo runs reliably in under three minutes.

25. Future Scope

- Live multi-camera CCTV integration
- Edge AI inference
- Multi-camera tracking and crowd-flow estimation
- LSTM/GRU/Transformer crowd forecasting
- Abnormal-event detection
- IoT resource sensors
- Volunteer mobile application
- GPS-based volunteer tracking
- Advanced VRP/linear-programming optimization
- Historical analytics
- Authorized integration with official emergency/operations systems

26. Final Product Positioning

This should be presented as an **AI-powered operational decision-support platform**, not simply a crowd-counting dashboard.

VARI SMART OPERATIONS

Crowd + Mobility + Volunteers + Resources + Sanitation

↓

Deep Learning & Decision Engine

↓

SENSE → PREDICT → OPTIMIZE → ACT

One-line pitch: “Vari Smart Operations is an AI-powered command platform that uses deep learning and real-time analytics to monitor crowd movement, predict operational risks, coordinate volunteers, and optimize essential resources during the Vari.”